Object-Oriented Programming with Java: Polymorphism & Interfaces

1. Abstract Base Classes and Inheritance Hierarchies

In Java, abstraction allows decoupling contract definitions from concrete execution logic using interfaces and abstract classes.

```
package edu.podsiksha.shapes;

public abstract class GeometricShape {
    protected String color;

    public GeometricShape(String color) {
        this.color = color;
    }

    public abstract double calculateArea();
    public abstract double calculatePerimeter();
}

public class Circle extends GeometricShape {
    private double radius;

    public Circle(String color, double radius) {
        super(color);
        this.radius = radius;
    }

    @Override
    public double calculateArea() {
        return Math.PI * radius * radius;
    }

    @Override
    public double calculatePerimeter() {
        return 2 * Math.PI * radius;
    }
}
```